

R E V I E W

A decade of photon-counting CT research: bibliometric trends and emerging themes

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ABSTRACT

Background and aim: Photon-counting computed tomography (PCCT) represents a significant advancement in medical imaging, offering improved spatial resolution, spectral imaging capabilities, and enhanced tissue characterization. This bibliometric analysis provides a comprehensive overview of PCCT research trends.

Methods: A total of 1,390 articles were retrieved from the Scopus database without time restrictions. Articles were ranked based on citation count, and data extraction included title, authorship, publication year, journal impact, citation metrics, study design, funding sources, and subject classification.

Results: Results indicate a rapid growth in PCCT-related publications, with the highest output recorded in 2024. The United States, Germany, and Switzerland emerged as leading contributors. Original research articles dominated (62.37%), followed by conference papers (14.68%) and review articles (12.23%). Keyword co-occurrence analysis highlighted major research themes, including spectral imaging, material decomposition, image quality, and artificial intelligence applications. The citation analysis revealed that foundational research on PCCT technology, image reconstruction, and artifact reduction has shaped the field.

Conclusions: The study underscores the growing clinical relevance of PCCT, particularly in cardiovascular imaging, oncology, and musculoskeletal applications. (www.actabiomedica.it)

Key words: photon-counting CT, bibliometric analysis, spectral imaging, citation analysis, medical imaging trends



Received: 24 January 2026 | Accepted: 9 March 2026

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Introduction

Photon-counting computed tomography (PCCT) is a type of X-ray computed tomography (CT) that utilizes a photon-counting detector (PCD) to detect X-rays by registering individual photon interactions. By tracking the energy deposited in each interaction, the PCD detector pixels record an approximate energy spectrum, making it a spectral or energy-resolved CT technique. In contrast, traditional CT scanners rely on energy-integrating detectors (EIDs), which capture the total energy deposited in a pixel, including a large number of photons and electronic noise. EIDs, therefore, only record photon intensity, similar to black-and-white photography, whereas PCDs also capture spectral information, much like color photography (1).

The first commercial PCCT scanner, introduced by Siemens, was approved by the U.S. Food & Drug Administration (FDA) in 2021 (2).

PCCT has the potential to overcome several limitations of existing CT technology. By counting incoming photons and measuring their energy, this technique improves the contrast-to-noise ratio and enhances spectral imaging. PCCT can reduce radiation exposure by at least 30%–40% of radiation dose (3,4). It also offers higher spatial resolution, which has already proven useful in cardiovascular CT and is expected to be beneficial in CT of the temporal bone, chest, breast, head and neck, orthopedic trauma, and potentially other applications (5–9). Additionally, it corrects beam-hardening artifacts and reduces blooming artifacts, improving luminal vessel evaluation (10). PCCT can enhance contrast with current iodinated contrast agents or lower iodine dosage when necessary (11). It also provides the opportunity to use alternative contrast agents, such as gadolinium, gold, and platinum, which could benefit patients with iodine allergies (12). Lastly, CT numbers produced by current CT scanners are influenced by the acquisition protocol and surrounding anatomy; with PCCT, each pixel will provide precise physical material and/or tissue information, enabling more accurate tissue characterization and enhanced perfusion imaging at lower radiation doses (13). In recent years, there has been a growing interest in PCCT due to its ability to overcome several limitations of conventional CT technology. The

introduction of the first commercial PCCT system and the ongoing development of additional models by other manufacturers highlight the increasing recognition of its clinical potential. As research in this field continues to expand, bibliometric analysis can provide valuable insights into emerging trends, key topics, and gaps that require further investigation. To date, no comprehensive bibliometric analysis has been conducted specifically on PCCT, underscoring the need for a systematic evaluation of the most influential publications and the current state of research in this rapidly evolving field.

Materials and methods

The bibliometric analysis was conducted on January 29, 2025. Ethical approval was not required for this study, as it is a retrospective bibliometric analysis that does not involve human subjects. Journal articles were identified using the Scopus database. The following search strategy was applied: “photon counting CT” OR “photon-counting computed tomography” OR “photon counting detector CT” OR “photon-counting detector computed tomography” OR “spectral photon counting CT” OR “photon-counting detector CT” OR “PCCT” OR “photon counting x-ray CT” OR “quantum counting CT” OR “quantum counting computed tomography”. No time or language restrictions were set, and both original research and review articles were included. To ensure comprehensive literature coverage, studies from fields beyond medicine, as well as those involving nonhuman subjects, were also considered.

The search yielded a total of 1,390 articles (Figure 1), which were ranked in descending order based on citation count. To ensure the relevance of the included studies, all retrieved articles were manually screened to confirm that PCCT was a primary focus, excluding papers that only briefly mentioned PCCT in the context of dual-energy CT. Articles were included if their title, abstract, or keywords contained the specified search terms. In cases where multiple articles had the same citation count, ranking was determined by the average number of citations per year, prioritizing more recent publications. For each article,

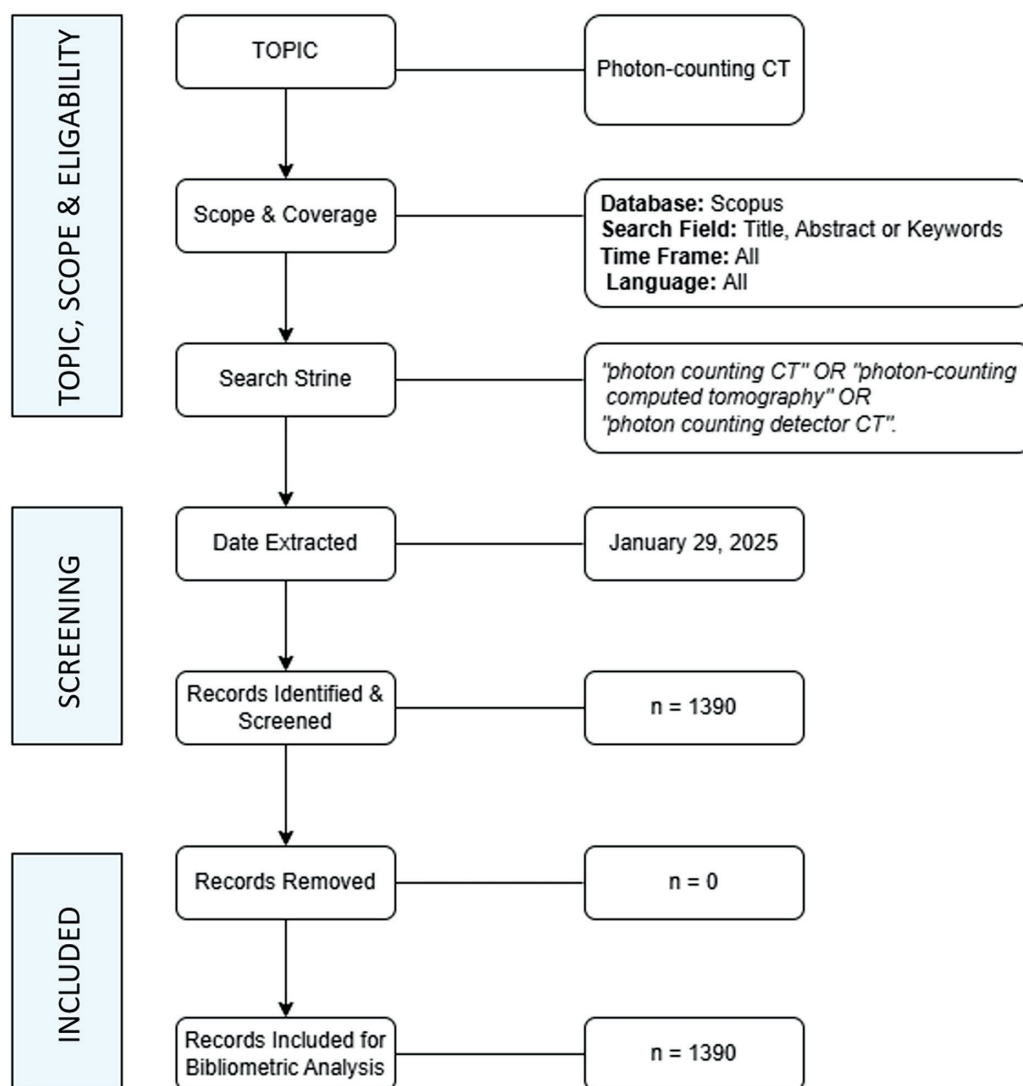


Figure 1. Flow diagram of the search strategy.

the following data were extracted: title, authors, year of publication, journal name, author affiliations, country of origin, article type, keywords, journal impact factor, total citations, mean citations per year, study design, funding status, and subject area. For articles with multiple authors from different institutions or countries, the first author's institution and country were recorded. Articles were categorized into the following types: original article, conference paper, review, editorial, book chapter, letter, conference review, book, and short survey.

Journal Cite-Scores were retrieved from their respective official websites.

Data analysis

Tables and charts were created using Microsoft Word and Excel. VOSviewer (version 1.6.120) was used for bibliometric analysis, facilitating the extraction of key information from numerous publications. It is commonly employed to construct collaboration, co-citation, and co-occurrence networks (14). In this

study, VOSviewer was used primarily for country analysis and keyword co-occurrence analysis. In the visual representation, nodes represent elements such as countries or keywords, with their size and color indicating frequency and classification. The thickness of connecting lines reflects the degree of collaboration or co-citation. Harzing's Publish or Perish 8.17.4863 software was used to calculate citation metrics.

Results

Description of retrieved literature

A total of 1,390 documents were identified from the Scopus database. Table 1 presents the distribution of document types. Original research articles comprised the majority (867/1,390, 62.37%), followed by conference papers (204/1,390, 14.68%), review articles (170/1,390, 12.23%), and editorials (54/1,390, 3.88%).

The retrieved documents collectively received 18,855 citations, averaging 942.75 citations per year and 13.57 citations per paper. The h-index of the dataset was 63. Most documents were published in English (1,365/1,390, 98.20%), with smaller contributions in German (12/1,390, 0.86%) and Chinese (7/1,390, 0.50%).

Regarding subject areas, 1,111 documents were classified under Medicine, 310 under Physics and

Astronomy, and 211 under Material Science (Supplementary Figure 1).

Growth of publications

Analyzing publication trends by year provides insights into the research field's growth and popularity over time (15). The highest number of publications was recorded in 2024, with 499 documents, whereas the lowest was in 2006, one year after technology's invention, with just a single publication. The study period demonstrated an exponential increase in publication volume (Figure 2). The highest citation rate per publication was observed for documents published in 2012 (averaging 60.08 citations per 12 publications), while the lowest was for those from 2024 and 2025, given the limited time since their release (Supplementary Table 1).

Keyword co-occurrence analysis

Using VOSviewer, a keyword co-occurrence mapping (minimum 10 occurrences) was performed. The most frequent keywords (excluding core search terms) included *spectral CT* (79 occurrences, 182 total link strength), *material decomposition* (87 occurrences, 159 total link strength), *image quality* (62 occurrences, 142 total link strength), *dual-energy CT* (51 occurrences, 101 total link strength), *spectral imaging* (34 occurrences, 71 total link strength),

Table 1. Types of retrieved documents.

Document Type	Total Publications (TP)	Percentage (%)
Article	867	62.37
Conference Paper	204	14.68
Review	170	12.23
Editorial	54	3.88
Book Chapter	23	1.65
Letter	20	1.44
Conference Review	7	0.50
Book	2	0.15
Short Survey	2	0.15
Other	41	2.95
Total	1390	100.00

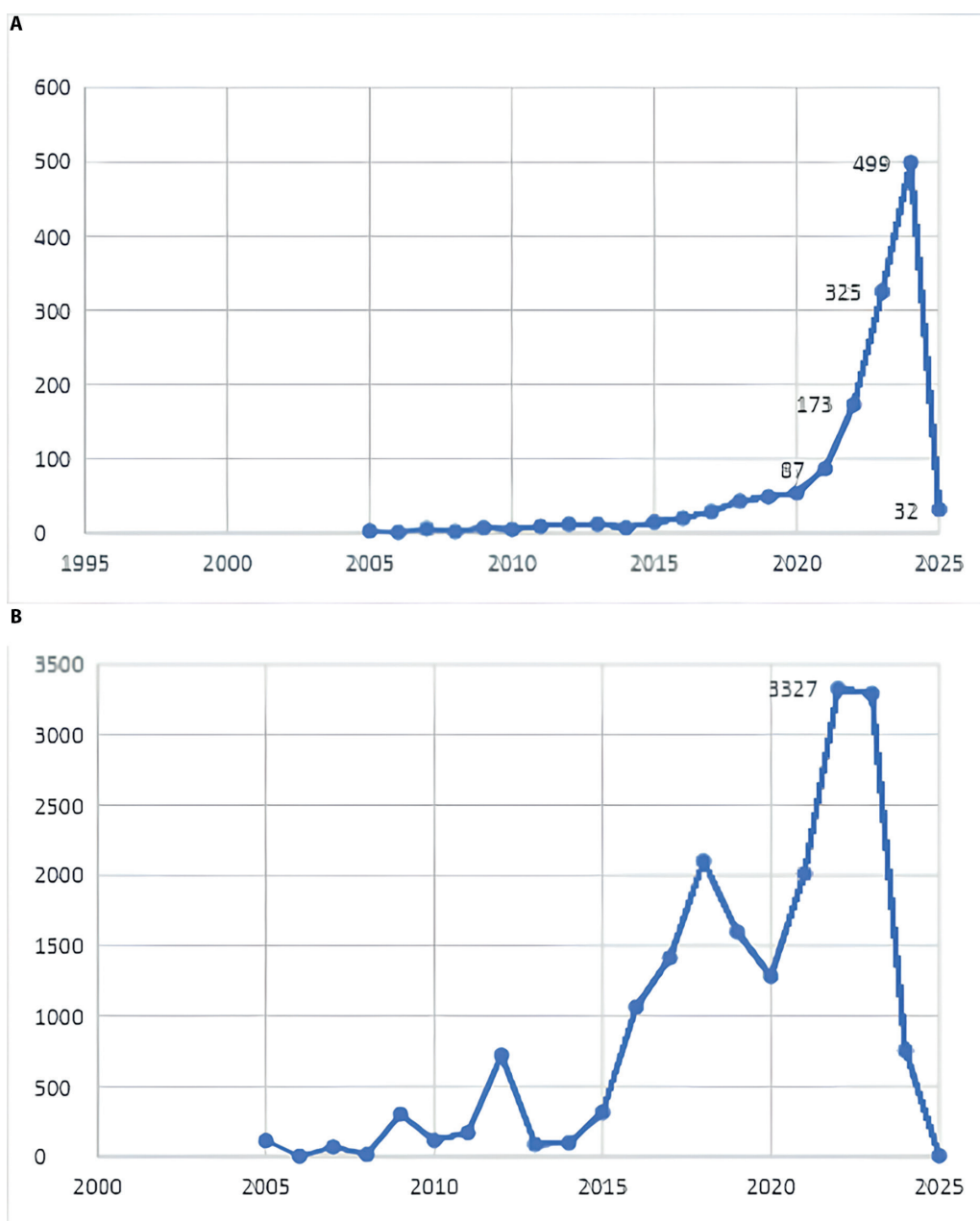


Figure 2. Trend of publications (A) and citations (B).

radiation dose (28 occurrences, 65 total link strength), *contrast media* (23 occurrences, 59 total link strength), *virtual monoenergetic imaging* (31 occurrences, 59 total link strength), *artificial intelligence* (28 occurrences, 52 total link strength), and *coronary artery disease* (33 occurrences, 51 total link strength) (Figure 3). Color clusters in the visualization indicate thematic subfields within PCCT

research. For example, Cluster 1 (red, 22 items) is associated with *PCCT applications*, including AI, diagnostic imaging, and contrast media. Cluster 2 (light blue, 11 items) focuses on *PCCT technology*, encompassing contrast agents, image reconstruction, and spectral imaging. Cluster 3 (blue, 11 items) relates to *cardiac imaging*, featuring keywords such as calcium, CAD, and heart disease.

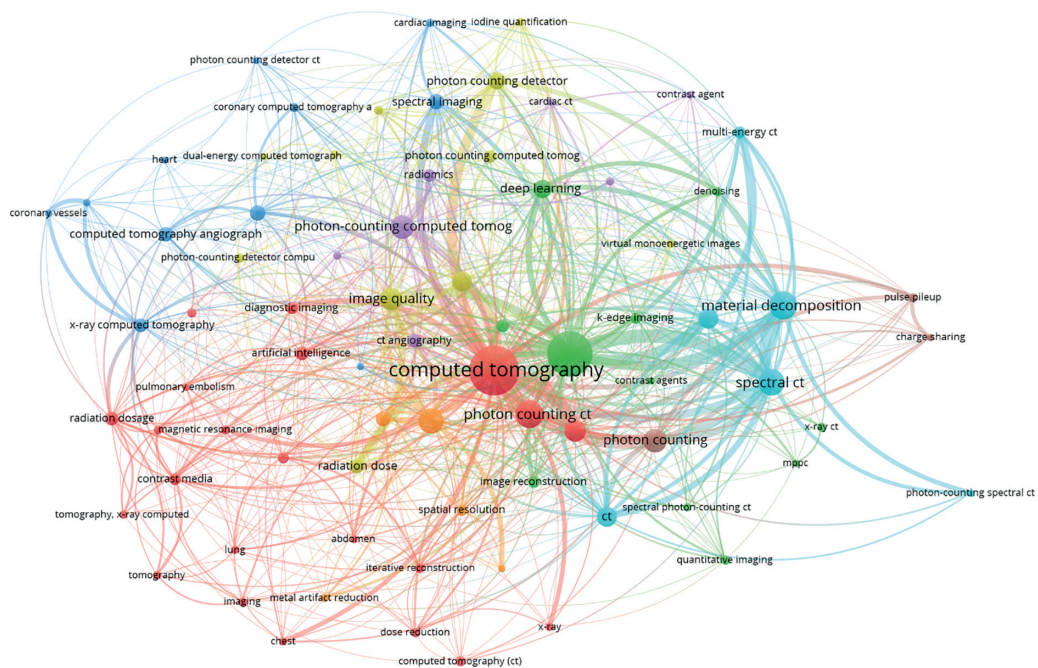


Figure 3. Network visualization map of the author keywords.

Authorship patterns and prolific authors

From 2015 onward, 158 authors published at least 10 articles on PCCT. Supplementary Table 2 lists the top 10 most prolific authors. The most productive was Leng, S. (88 publications), followed by McCollough, C.H. (81 publications), both affiliated with the Department of Radiology of Mayo Clinic, US and Siemens Healthcare.

Geographical distribution of publications

Researchers from 54 countries contributed to the dataset. A choropleth map illustrating each country's contribution is shown in Figure 4. The United States ranked first with 588 publications (42.30%), followed by Germany (28.71%), Switzerland (8.42%), France (6.83%), Japan (6.76%), China (6.62%), and the Netherlands (5.32%). Supplementary Figure 2 presents a network visualization of international collaborations (minimum 5 publications per country).

Preferred journals

Table 2 lists the top 10 journals in PCCT research. *Progress in Biomedical Optics and Imaging – Proceedings*

of SPIE led with 138 publications, followed by *Medical Physics* (75), *Investigative Radiology* (69), *Radiology* (62), *European Radiology* (55), *European Journal of Radiology* (52), and *Diagnostics* (50). Radiology had the highest citation count (2,555), followed by *Investigative Radiology* (2,392) and *European Radiology* (1,269). The highest citations-per-paper ratio was observed in *Radiology* (41.21), followed by *Investigative Radiology* (34.67).

Funding sponsors

The *National Institutes of Health (US)* funded 188 articles, followed by the *U.S. Department of Health and Human Services* (97), *National Institute of Biomedical Imaging and Bioengineering (US)* (69), *Siemens* (63), and *National Natural Science Foundation of China* (50) (Supplementary Figure 3).

Top cited documents

Table 3 lists the 20 most-cited articles. Among these, seven were review articles, including the top two cited papers, and thirteen were original articles. The most-cited article was *Photon-counting CT: Technical principles*

Table 2. Top 10 publishing journals on PCCT research.

Source Title	TP	TC	C/P	Publisher	Cite Score
Progress In Biomedical Optics And Imaging Proceedings Of SPIE	138	649	4.70	The International Society for Optics and Photonics	6.4
Medical Physics	75	909	12.12	Wiley	6.7
Investigative Radiology	69	2392	34.67	Wolters Kluwer	15.1
Radiology	62	2555	41.21	Radiological Society of North America (RSNA)	35.2
European Radiology	55	1269	23.07	Springer	9.4
European Journal Of Radiology	52	500	9.62	Elsevier	6.7
Diagnostics	50	580	11.60	MDPI (Multidisciplinary Digital Publishing Institute)	4.7
Physics In Medicine And Biology	50	1280	25.6	Institute of Physics Publishing	6.5
American Journal Of Roentgenology	36	299	8.31	American Roentgen Ray Society (ARRS)	8.6
Proceedings Of SPIE The International Society For Optical Engineering	35	64	1.83	The International Society for Optics and Photonics	2.7

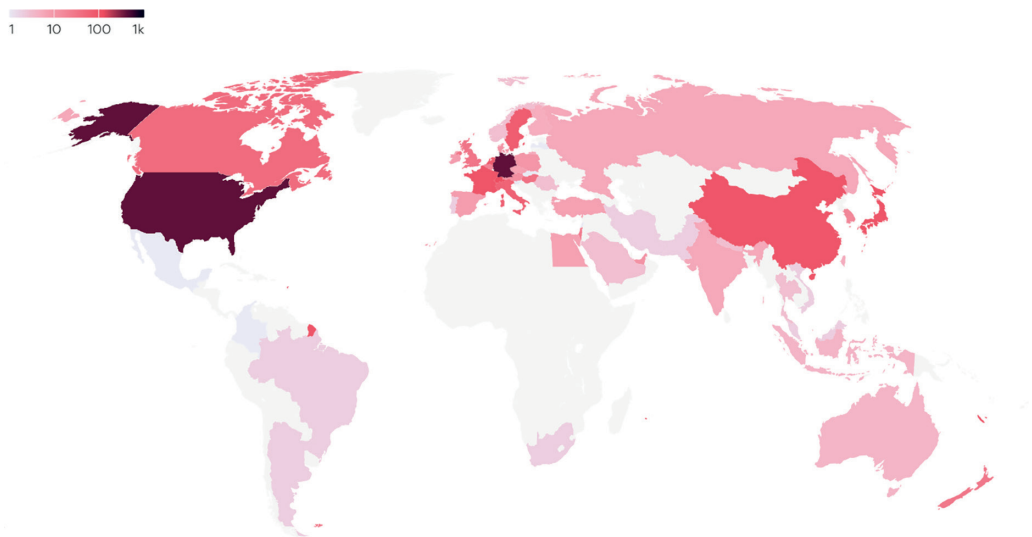


Figure 4. Choropleth Map of each country's contribution in PCCT research.

and clinical prospects (1), published in *Radiology* (2018), with 830 citations. This article has received an average of 118.57 citations per year, making it the most frequently cited PCCT article on an annual basis. The second most-cited article, *CT artifacts: Causes and reduction techniques*

(16), published in *Imaging in Medicine* (2012), had 649 citations, and the second-highest citations-per-year article was *First Clinical Photon-counting Detector CT System: Technical Evaluation* (17), published in *Radiology* (2022), with 106.00 citations per year.

Table 3. Characteristics of the top 20 most-cited articles on PCCT.

Rank	No. of citations	Citations per year	Year of publication	Type	Journal	Title
1	830	118.57	2018	Review	Radiology	Photon-counting CT: Technical principles and clinical prospects (1)
2	649	49.92	2012	Review	Imaging in Medicine	CT artifacts: Causes and reduction techniques (16)
3	384	64.00	2019	Review	European Radiology	The evolution of image reconstruction for CT—from filtered back projection to artificial intelligence (40)
4	354	59.00	2019	Review	Radiographics	Photon-counting detector CT: System design and clinical applications of an emerging technology (41)
5	343	68.60	2020	Review	Physica Medica	Photon-counting CT review (42)
6	318	106.00	2022	Article	Radiology	First Clinical Photon-counting Detector CT System: Technical Evaluation (17)
7	266	16.63	2009	Article	IEEE Transactions on Nuclear Science	Photon counting energy dispersive detector arrays for X-ray imaging (43)
8	239	26.56	2016	Article	Investigative Radiology	Human imaging with photon counting-based computed tomography at clinical dose levels: Contrast-to-noise ratio and cadaver studies (44)
9	206	22.89	2016	Article	Physics in Medicine and Biology	Evaluation of conventional imaging performance in a research whole-body CT system with a photon-counting detector array (45)
10	182	22.75	2017	Article	Medical Physics	Photon-counting CT for simultaneous imaging of multiple contrast agents in the abdomen: An in vivo study (46)
11	177	59.00	2022	Article	Radiology	Coronary CT Angiography with Photon-counting CT: First-In-Human Results (47)
12	177	19.67	2016	Article	Radiology	Abdominal imaging with contrast-enhanced photon-counting CT: First human experience (48)
13	176	44.00	2021	Review	Physics in Medicine and Biology	Photon-counting x-ray detectors for CT (49)

Rank	No. of citations	Citations per year	Year of publication	Type	Journal	Title
14	174	29.00	2019	Article	Scientific Reports	Effect of Gold Nanoparticle Size on Their Properties as Contrast Agents for Computed Tomography (50)
15	136	19.43	2018	Article	Investigative Radiology	Photon-Counting Computed Tomography for Vascular Imaging of the Head and Neck: First in Vivo Human Results (51)
16	119	14.88	2017	Article	Scientific Reports	Multicolor spectral photon-counting computed tomography: In vivo dual contrast imaging with a high count rate scanner (52)
17	117	14.63	2017	Article	Radiology	Spectral Photon-counting cT: Initial Experience with Dual-Contrast Agent K-Edge Colonography (53)
18	113	56.50	2023	Review	Diagnostic and Interventional Imaging	Spectral CT imaging: Technical principles of dual-energy CT and multi-energy photon-counting CT (54)
19	110	15.71	2018	Article	Investigative Radiology	Photon-Counting CT: High-Resolution Imaging of Coronary Stents (55)
20	106	13.25	2017	Article	Physics in Medicine and Biology	Spectral performance of a whole-body research photon counting detector CT: Quantitative accuracy in derived image sets (56)

Across the top 20 most-cited articles, the average total citations were 258.8 (range 106–830), with an average of 42.05 citations per year (range 13.25–118.57). Twelve different journals contributed to the top-cited list, with *Radiology* leading (five publications, 25%).

Discussion

PCCT represents a transformative advancement in the field of diagnostic imaging, addressing several limitations of conventional CT technology (1). The bibliometric analysis of PCCT literature provides valuable insights into the evolving research landscape, highlighting key trends, prolific authors, influential

institutions, and primary areas of focus (15). To date, no comprehensive bibliometric analysis has examined the full scope of PCCT, underscoring the need for a systematic evaluation of its most influential publications and overall research landscape. A recent study has addressed cardiac PCCT specifically, but the broader field has not yet been explored from a bibliometric perspective (18). The present analysis delineates three central dimensions of the current PCCT research landscape: exponential growth in publication volume, structural concentration of contributions within selected countries and institutions, and the emergence of artificial intelligence and cardiovascular imaging as predominant areas of investigation. The exponential increase in annual publications, culminating in a peak

in 2024, reflects the rapid consolidation of PCCT as an active research domain. This trajectory suggests a transition from early-stage technological development to broader clinical dissemination. The temporal proximity between the 2021 FDA approval of the first commercial PCCT system and the subsequent surge in publications strongly suggests that regulatory clearance and scanner commercialization acted as pivotal drivers of research activity (19).

Earlier publications accumulated high total citation counts, reflecting their foundational role in establishing the physical and technical principles of photon-counting imaging (1). In contrast, studies published between 2021 and 2024 demonstrate elevated annual citation rates, particularly those focused on clinical implementation. (20,21).

This shift in citation dynamics suggests a transition from theory-oriented and feasibility research toward application-driven validation studies. The field therefore appears to be moving from technological proof-of-concept toward performance assessment in clinical environments (22).

The bibliometric distribution reveals a pronounced concentration of research output in the United States and Germany, followed by Switzerland and other European and Asian countries. This pattern likely reflects early access to commercial PCCT systems and the presence of strong academic-industry collaborations. Institutions with established partnerships with technology vendors appear to occupy central positions within the publication network, underscoring the translational structure of PCCT research. Funding patterns further reinforce this observation, with substantial contributions from governmental agencies and industry sponsors. These findings suggest that the development and validation of PCCT are closely linked to resource-intensive infrastructures and collaborative ecosystems. The geographic concentration of output may also explain the rapid expansion observed in 2024. As scanner availability increased within high-resource academic centers, publication volume rose correspondingly. Conversely, the relative underrepresentation of other regions suggests that limited access to hardware and infrastructural constraints may currently restrict broader global dissemination.

Keyword co-occurrence analysis identifies AI and cardiovascular imaging as the most prominent and rapidly expanding research clusters in PCCT. The frequent appearance of “coronary artery disease” among high-occurrence terms highlights cardiovascular applications as a primary domain for clinical validation, reflecting both technological suitability and high clinical demand (23-25). AI, in parallel, represents a methodological cluster driven by the complexity of spectral datasets generated by photon-counting detectors. (26-28). The bibliometric prominence of these clusters indicates a strategic focus of the field: cardiovascular imaging provides a robust platform to demonstrate PCCT advantages, including high spatial resolution, spectral discrimination, and artifact reduction, while AI enables advanced image reconstruction, quantitative analysis, and automated interpretation. (24,29,30). Together, these clusters illustrate a shift in research priorities from purely technical investigations toward application-driven studies, where innovation is guided not only by detector design but also by computational approaches. Analysis of highly cited publications supports this trend. Early influential articles primarily consisted of technical reviews and conceptual frameworks, establishing the principles of photon-counting imaging (1,21). More recent highly cited studies, particularly those published between 2021 and 2024, focus on clinical implementation, including coronary CT angiography and high-resolution vascular imaging. (23,31). The elevated annual citation rates of these clinical studies suggest strong engagement from the radiological community and reflect an ongoing transition toward real-world validation of PCCT performance. (20,32). Although cardiovascular imaging and AI dominate the research landscape, other potential application domains remain comparatively underexplored. This uneven distribution likely reflects limited scanner accessibility, implementation costs, and the early stage of translational development in less-studied areas. Similarly, while dose reduction and contrast optimization are recurring themes, multicenter outcome studies and cost-effectiveness analyses have yet to form major research clusters. (33-39). These observations highlight both the current focus of PCCT research and opportunities for future expansion as adoption increases globally.

This bibliometric study has several limitations that should be acknowledged. The analysis was conducted exclusively using the Scopus database. Although Scopus provides extensive coverage and structured citation tracking, reliance on a single database may have influenced both the overall number of retrieved publications and the corresponding citation counts, given differences in indexing policies and journal inclusion across platforms.

Citation analysis is inherently time-dependent. Publications from 2024–2025 have had limited time to accumulate citations, creating an unavoidable temporal imbalance when compared with earlier studies. As a result, citation-based metrics may underestimate the scientific impact of the most recent contributions.

Country-level productivity was attributed according to the first author's institutional affiliation. While this approach is widely used in bibliometric research to ensure methodological consistency, it does not fully capture multinational collaborations. Although international co-authorship patterns are partially illustrated in the collaboration network (Supplementary Figure 2), assigning each publication to a single country may underestimate the contribution of partner institutions involved in multi-country research.

The dataset also included conference papers and abstracts. Their inclusion enables a broader representation of emerging research activity and technological development; however, these document types do not always undergo the same level of peer review as full-length journal articles. Consequently, their impact within citation-based analyses should be interpreted with caution.

Keyword co-occurrence analysis was performed using a minimum threshold of 10 occurrences to enhance visualization clarity and reduce noise. While this methodological choice improves the readability and interpretability of the network map, it may exclude less frequent but potentially relevant specialized or emerging terms, thereby underrepresenting niche areas within photon-counting CT research.

Conclusion

This bibliometric analysis provides a comprehensive overview of PCCT research trends, highlighting the rapid expansion of the field, key contributors, and

influential publications. The increasing number of publications, diverse international contributions, and high-impact research indicate the growing importance of PCCT in modern medical imaging. Future research should focus on addressing existing knowledge gaps, optimizing clinical applications, and fostering collaborative efforts to further advance PCCT technology and its implementation in healthcare settings.

List of Abbreviations

PCCT: Photon-Counting Computed Tomography
 CT: Computed Tomography
 PCD: Photon-Counting Detector
 EID: Energy-Integrating Detector
 FDA: Food & Drug Administration
 CNR: Contrast-to-Noise Ratio
 VMI: Virtual Monoenergetic Imaging
 AI: Artificial Intelligence
 CAD: Coronary Artery Disease
 NIH: National Institutes of Health

Ethic Approval: Not applicable.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Declaration of Interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Authors' Contribution: "Conceptualization, G.L. and N.M.; methodology, C.M. and S.G.; software S.G. and I.V.; validation, G.L. and N.M.L.; formal analysis, E.C.; investigation, C.M.; resources, E.C.; data curation, I.V.; writing—original draft preparation, E.C. and G.L.; writing—review and editing, N.M. and I.V.; visualization, C.M. and S.G.; supervision, N.M. and N.M.L.; project administration, G.L. and N.M.L.; funding acquisition, N.M. and N.M.L.; All authors have read and agreed to the published version of the manuscript."

Declaration on the Use of AI: N.A.

Acknowledgements: N.A.

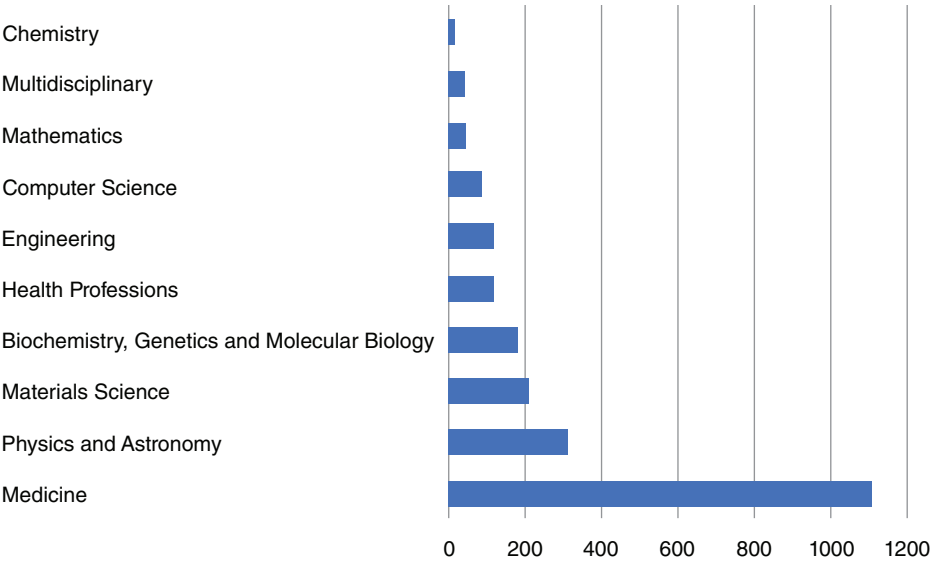
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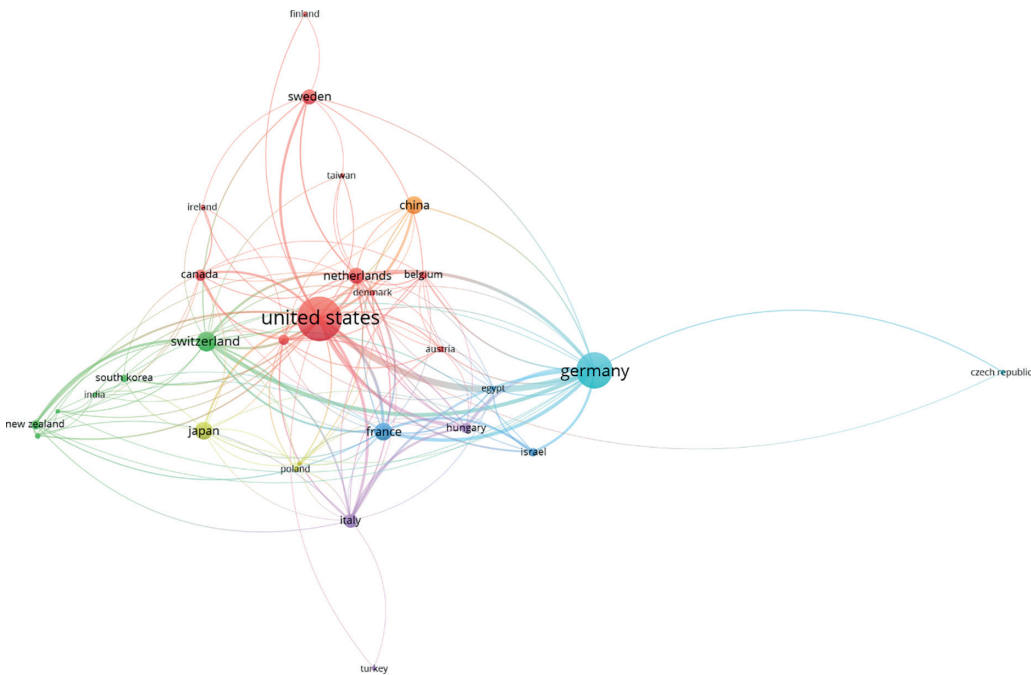
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Appendix

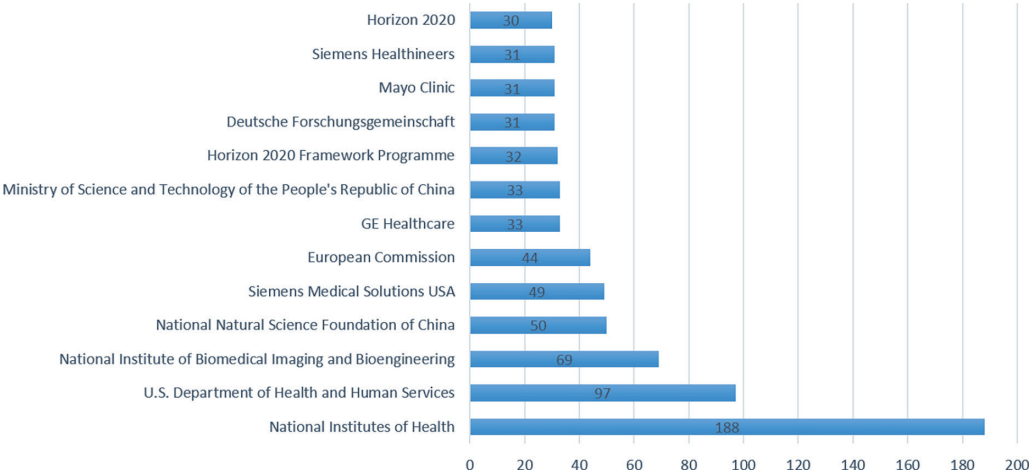
Supplementary files



Supplementary Figure 1. Top 10 Subject are



Supplementary Figure 2. Network visualization map of international collaboration among countries with minimum productivity of 5 documents. The thickness of the connecting line between any two countries indicates the strength of collaboration. Countries with similar colors form one cluster.



Supplementary Figure 3. Most influential funding sponsors with a minimum of 30 publications.

Supplementary Table 1. Annual number of publications and citation matrix.

Year	TP	TC	C/P	h-index	g-index
2025	32	8	0.22	2	2
2024	499	759	1.53	11	13
2023	325	3302	10.16	27	39
2022	173	3333	19.27	32	51
2021	87	2016	23.17	27	42
2020	54	1285	23.80	20	34
2019	49	1604	32.73	18	40
2018	43	2105	48.95	19	43
2017	29	1414	48.76	17	29
2016	20	1064	53.20	12	20
2015	15	314	20.93	10	15
2014	7	98	14.00	4	7
2013	12	87	7.25	3	9
2012	12	721	60.08	5	12
2011	9	171	19.00	5	9
2010	5	115	23.00	4	5
2009	7	301	43.00	3	7
2008	2	16	8.00	1	2
2007	6	69	11.50	3	6
2006	1	3	3.00	1	1
2005	3	114	38.00	2	3

Supplementary Table 2. Top 10 most productive authors.

Author	Affiliation	Country	TP
Leng, S.	Department of Radiology, Mayo Clinic and Siemens Healthcare	United States	88
McCollough, C.H.	Department of Radiology, Mayo Clinic and Siemens Healthcare	United States	81
Rajendran, K.	Department of Radiology, Mayo Clinic and Siemens Healthcare	United States	57
Alkadhi, H.	Institute of Diagnostic and Interventional Radiology, University Hospital Zurich	Switzerland	55
Schmidt, B.	Siemens Healthcare GmbH, Computed Tomography	Germany	55
Samei, E.	Department of Radiology, Duke University Medical Center	United States	50
Yu, L.	Department of Radiology, Mayo Clinic and Siemens Healthcare	United States	45
Mergen, V.	Institute of Diagnostic and Interventional Radiology, University Hospital Zurich	Switzerland	40
Douek, P.	University Lyon	France	38
Emrich, T.	Siemens Medical Solutions	United States	37

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